



Fish traders in artisanal fisheries on the Kenyan coast

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Abstract

Purpose – Marine fisheries are one of the few economic activities present everywhere along the Kenyan coastline. The local population is involved mainly in artisanal fishing which uses small non-motorized fishing crafts that stay close to shore. Some of the catch is destined for local consumption but most is for sale. The purpose of this paper is to question whether fish traders in artisanal fisheries along the Kenyan coast earn enough money from only fish trading to support a household.

Design/methodology/approach – Fish traders were surveyed at two landing sites at each of five coastal tracts. Structured questionnaires, informal interviews and participatory observations were used in collecting data.

Findings – Average income for the fish traders from only fish trading was Ksh 1,268 per week; only 20.3 percent of the households was at or above the poverty line. However, there was a large difference between male and female traders in earning. Men earned Ksh 1,693 per week and women Ksh 795 per week. The poverty line for households was reached by 30.8 percent of the male traders but only by 8.8 percent of the female traders.

Originality/value – Livelihood diversification could greatly help improve the income. It was estimated that when earnings other than from fish trading (from the traders or someone else in the household) were added to that of fish trading, 27.4 percent of the households was at or above the poverty line. For men traders, it was 54 percent of the households but for women it was only 15 percent.

Keywords Kenya, Fishing, Coastal regions, Trade, Gender

Paper type Research paper



Introduction

The total number of full-time, part-time and seasonal fishers in Africa in 2005 was estimated at 2.73 million, of whom more than 90 percent belonged to the artisanal sector (FAO, 2007). Including the people involved in the processing and other related industries, and taking into account their families and dependants, this may increase the number of people relying on fisheries for income to 50-60 million (Heck *et al.*, 2007). Artisanal fishery refers to a small-scale enterprise that involves households, as opposed to companies, and in which there is relatively little capital input. Artisanal fishers typically make short trips in small, non-motorized fishing crafts and stay within three nautical miles of shore (FAO, 2004; Ochiewo, 2004). On the Kenyan

coast, 70 percent of the fishing vessels are canoes and *daus*, which are suited mainly for reef and in-reef fishing (Hoorweg *et al.*, 2009a). The catch is destined for local consumption and for sale, and thus provides food, employment and financial gains (Walmsley *et al.*, 2006; Abbott *et al.*, 2007).

Artisanal fisheries are characterized by uncertainty and depend strongly on seasons. For example, along the East African coast, fish catches are poorest from April to July, during the *kusi* season, when southeast monsoon winds occur. This period is characterized by heavy rainfall and rough seas. During this time, fishers go out less to sea and tend to avoid the out-of-reef areas and deep waters. In contrast, fish catches are best from August to March, during the *kaskazi* season, when northeast monsoon winds occur. Seas are relatively calm allowing fishers to go out more often and venture out further (Ochiewo, 2004; Hoorweg *et al.*, 2009a).

In general, artisanal fishers are poorly equipped for the dangers of the sea, often dependent on intermediaries and ship owners and have to deal with fluctuating incomes (Charles, 2001). Their gear has to be adapted and different fishing techniques have to be mastered to match the seasonal changes and the large numbers of species. Generally, fishers are uneducated with a low literacy rate (Branch *et al.*, 2002; Obura *et al.*, 2002), are poorer than their non-fisher neighbours (Cinner *et al.*, 2010) and are politically underrepresented because they live in marginal and remote areas and because of their regular absences due to long periods at sea (Acheson, 1981).

Kenyan artisanal fishing

Kenya belongs to the bottom 20 percent of countries in the world in terms of economic and human development. About half the population is termed poor by national standards and the poorest regions of the country are North-Eastern, Rift Valley and Coast Province, with about 64 percent of the latter below the poverty line[1] (Kenya, 2001). The reasons for poverty in Coast Province include unfavourable climate, poor agro-ecological conditions, lack of employment opportunities and low level of education (Hoorweg *et al.*, 2000). Marine fisheries are one of the few economic activities present everywhere along the coastline and the number of fishers is steadily increasing (Hoorweg *et al.*, 2009a).

While off-shore fishing is the domain of a few Kenyan vessels and some visiting factory trawlers from foreign countries, the local population is involved mainly in artisanal fisheries. The marine landings in 2003 were about 7,616 metric tons with an estimated value to the fishers of Ksh 587.2 million or US\$8.1 million (Kenya, 2007). It was reported that the artisanal fish harvest from coral reefs accounted for over 90 percent of the entire marine fish catch (KMFRI, 2002).

Marine fisheries along the Kenyan coast employed an estimated 10-12,000 fishers in 1999. Including workers in the support industry and household dependents, some 167-200,000 people out of a total of 2.49 million people (~7.5 percent of the population) were wholly or partly dependent on fisheries. Fishing is traditionally a male profession among artisanal fishers along the Kenyan coast; however, women are quite active in fish processing and trading (Wangila *et al.*, 2007). This is also the case in many other areas of the world as, for example, in West Africa, where women often control the processing and marketing sectors in small-scale fisheries (Bennett, 2005). We studied fish traders in artisanal fisheries along the Kenyan coast and questioned whether they earned enough money from only fish trading to support a household.

Methods

Study site – Kenyan coast

The East African coast stretches for 5,500 km and includes the coastlines of Somalia, Kenya, Tanzania and Mozambique. The Kenyan coast covers about 600 km from the Somali border in the north to the Tanzanian border in the south (Frazier, 1993; UNEP, 1998a). The climate on the Kenyan coast is dominated by large-scale pressure systems from the Western Indian Ocean and two distinct monsoon periods. The northeast monsoon prevails from November to March, the southeast monsoon from May to October. Annual rainfall on land close to the sea is about 1,000 mm whereas a few kilometres inland it decreases to 700 mm. The shoreline exists of rocky fossil coral cliffs, mangrove stands and sandy beaches (UNEP, 1998a; Obura, 2001). A fringing reef parallels the shores to the south and to the north anywhere from 0.1 to 1.0 km offshore (Figure 1). Coral reefs are among the most productive ecosystems with great biodiversity (McClanahan and Obura, 1996). The reefs also contain special habitats like tide pools and, with shorelines, are important fishing grounds for the artisanal fishers (Aloo, 2000).

Coast Province has six administrative districts that border the sea, namely from north to south: Lamu, Tana River, Malindi, Kilifi, Mombasa and Kwale. The study area was situated in Malindi and Kilifi districts, extending from Ras Ngomeni (the Ngomeni peninsula) to Takaungu Creek, a coastline of roughly 125 km and consists of five coastal tracts with two landing sites in each. From north to south the tracts are Ngomeni, Malindi, Mida, Kilifi and Takaungu, which, more or less, covers the coast of Kilifi and Malindi districts, excepting about 25 km north and 25 km south (Figure 1).

Survey

The survey on traders covered two landing sites at the five coastal tracts for a total of ten sites and was done between December 1999 and March 2001. Structured questionnaires, informal interviews and observations were used in collecting data. At each landing site, traders were randomly selected and interviewed either on-site or at their homes. In total, 186 traders were interviewed at the coast tracts as follows: Ngomeni ($n = 32$), Malindi ($n = 42$), Mida ($n = 37$), Kilifi ($n = 43$) and Takaungu ($n = 32$). The data collected included the composition of fish harvested, volumes caught and buying and selling prices. We also gathered information on the storage, transport and destination of the fish as well as socio-economic household characteristics.

Population and ethnic groups

In 1999, the total population of Coast Province was about 2.49 million people and that of Malindi and Kilifi districts was close to 826,000 people (Kenya, 2002; STATOIDS, 2008). Coastal inhabitants are generally poor and rely heavily on simple, traditional methods of food production, which include subsistence farming and artisanal fishing. Fishing provides the principal source of animal protein and is an income earning substitute, especially during the high fishing season. The predominant near-shore activities include artisanal fishing, shrimp trawling, other commercial extraction and tourism, but marine resource use is largely unregulated (Obura, 2000).

The coastal inhabitants belong to various ethnic groups, mainly *Mijikenda*, *Bajun* and *Swahili*. In the five study tracts, *Mijikenda* was the main group and comprised

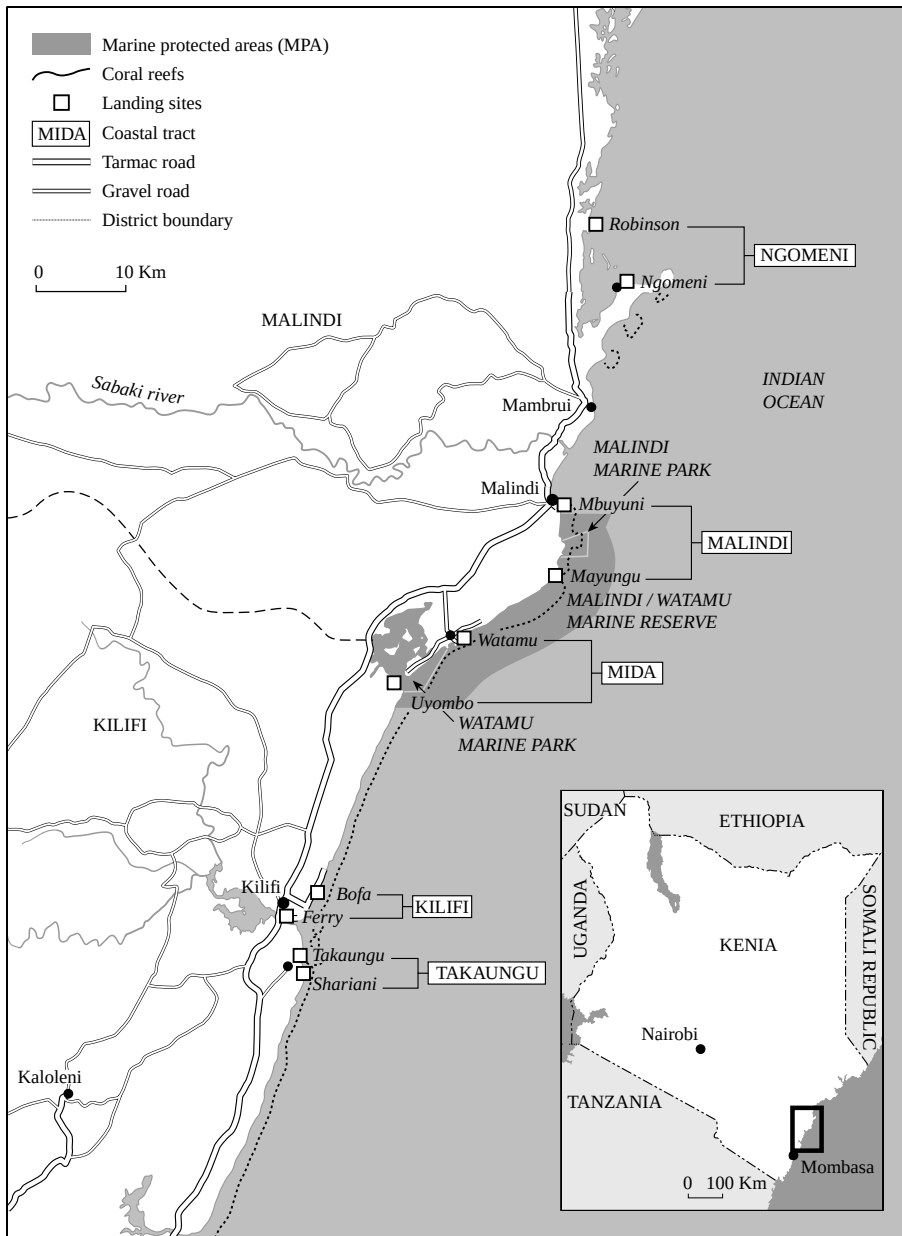


Figure 1. Map of the Kenyan coast including the Malindi and Kilifi districts, the coastal tracts and the landing sites

about 60 percent of the study population. They were traditionally from a farming background and lived on the coastal plateau and coastal range. Over the last century and a half, many *Mijikenda* have moved into the coastal strip and have joined the fisheries in large numbers since the 1960s, although this ethnic group does not have

a tradition of sea fishing, has little traditional knowledge on how to manage marine resources and does not provide apprenticeships for young fishers (Glaesel, 1997). In fact, during the study, few of the *Mijikenda* were second-generation fishers. They are mostly Christians (43 percent) but some are Muslims (29 percent) or practice African traditional religion (27 percent). The *Bajun* and *Swahili* are coast dwellers of long tradition. The *Bajun* are fishers *par excellence* and have relied on fishing or fishing related activities for generations. Half the *Bajun* fishers during the study came from fisher families. There are also some *Swahili* fishers although their numbers are small. The *Bajun* and *Swahili* are all Muslims.

Results

About half the traders were women. The highest percentage of female traders at a site was 64 percent, recorded at Mida, whereas the lowest percentage was 23 percent, recorded at the Kilifi tract. The majority of traders were of *Mijikenda* extraction (69.4 percent) and far fewer were of *Bajun* origin (22.6 percent). There were almost no traders of *Bajun* origin in the Kilifi and Takaungu coastal tracts (Table I).

Traders came from the landing sites or nearby villages (27.5 percent), the nearest villages (NVs) on the tarmac road (16.1 percent) and urban centres in the area (36.6 percent). Most traders were in the age groups of 20-29 years (42.1 percent) and 30-39 years (27.3 percent). A large number of traders had not attended any formal education (37.0 percent). There were only small differences between coast tracts in respect to age and education of the traders.

There were no fishers who fished only for subsistence needs. Fishers usually sold most of the catch and took some fish home for consumption. The part of the catch that was taken home, locally referred to as *kitoweo*, has been estimated to account for 4 percent of the artisanal catches world-wide (UNEP, 1998a, b). Almost half the fishers kept some fish as bait for the next day. Two-thirds of the fishers sold the fish as is, in fresh condition. The percentage of fishers doing this was highest at Malindi and Kilifi, sites that have ready access to markets, followed by Mida and Takaungu. At Ngomeni, the most remote tract, two-thirds of the fish were gutted and dried.

Once the catch was landed, the fish were nearly always sold to traders and intermediaries, hardly ever directly to consumers. In about a third of the cases they were sold to a *tajiri*[2] who had the right of first refusal for the catch of certain fishers. This occurred more often in Ngomeni, Kilifi and Takaungu. Most men were middlemen while most women were fishmongers. About 88.2 percent of women traders sold directly to private consumers while only 28.3 percent of men traders did so (Table II).

Table I.
Ethnic groups of the male
(*n* = 93) and female
(*n* = 93) fish traders
along the Kenyan coast
of Malindi and Kilifi
districts

	Male	Female	Total
<i>Ethnic group</i>			
Arab/Swahili	10.8	0.0	5.4
Bajun	43.0	2.2	22.6
Mijikenda	41.9	96.8	69.4
Other	4.3	1.1	2.7
Note: Values are in percent			

Almost all marine fish marketing in Malindi and Kilifi districts was on a local scale which involved buying and selling at landing sites, open markets and fish shops. Most of the fish was sold fresh and consumed locally, although some dealers took the fish to Mombasa, Nairobi and elsewhere. The number of days per week that the traders purchased fish was quite similar in the *kusi* (low) and *kaskazi* (high) seasons, but the amount of fish purchased in the *kaskazi* season was much greater. In the *kaskazi* season, 41.8 percent of the men traders bought at least 100 kg fish daily and 19.8 percent bought at least 250 kg daily, but in the *kusi* season only 2.3 percent of the men bought over 100 kg daily and no men bought over 250 kg daily. None of the women traders bought 100 kg of fish daily in any season (Table III).

Traders usually had a specialty in which they traded; men traded mainly in medium and large fish while women traded mainly in small and medium fish. In addition, men traded more in crustaceans and octopus than women (Table IV). Traders differed

	Male	Female	Total
<i>Type of trader</i>			
Small scale	19.9	2.7	11.3
Fishmonger	33.8	84.8	59.3
Middleman	35.9	10.4	23.2
Large scale	10.4	2.1	6.2
<i>Next buyer</i>			
Private consumers	28.3	88.2	58.4
Hotels	5.4	0.0	2.7
Fish sellers	8.7	1.1	4.9
Middlemen	1.1	0.0	0.5
Combination	56.5	10.8	33.5

Note: Values are in percent

Table II.
Type of male ($n = 97$)
and female ($n = 93$) fish
traders and the identity of
next buyer from fish
traders along the Kenyan
coast of Malindi and Kilifi
districts

	Male	Low season Female	Total	Male	High season Female	Total
<i>Days per week</i>						
Sample size (n)	90	91	181	91	92	183
Up to 2	12.1	12.1	12.2	7.7	2.2	4.8
3-5	7.7	14.3	11.1	5.5	25.1	15.4
6	4.4	14.3	9.4	5.5	15.2	10.4
7	75.6	59.3	67.4	81.3	57.6	69.4
<i>Fish bought (kg)</i>						
Sample size (n)	86	82	168	91	91	182
0-4	5.8	59.8	32.1	2.2	27.5	14.8
5-9	17.4	25.6	21.4	5.5	30.8	18.1
10-19	29.1	13.4	21.4	9.9	26.4	18.1
20-49	33.7	1.2	17.9	19.8	13.2	16.5
50-99	11.6	0.0	6.0	20.9	2.2	11.5
100-249	2.3	0.0	1.2	22.0	0.0	11.0
250+	0.0	0.0	0.0	19.8	0.0	9.9

Note: Values are in percent of each category

Table III.
Buying frequency in days
per week and amount fish
bought in kg per day by
male and female fish
traders in the low (*kusi*)
and high (*kaskazi*)
seasons along the Kenyan
coast of Malindi and Kilifi
districts

significantly in the average amount of fish they purchased daily, roughly a third bought < 10 kg, a third between 10 and 50 kg and a third more than 50 kg. On average, traders had 7.3 years of experience in the trade with half the traders having more than five years. Most traders frequented only one landing site while almost 40 percent of the traders frequented two, three or even more sites (Table V). Only a third of the traders offered incentives such as financial assistance (15 percent) or foodstuffs (9 percent) during times of hardship. Fishmongers, which comprised almost half the traders, fried the fish before selling them. Only a fifth of the traders had access to a cooler or freezer (Table VI). The destination of the fish included the NV (29 percent of the traders), the NV on the tarmac road (27 percent) and urban destinations such as Malindi, Kilifi and Mombasa. Transport was either on foot (43 percent), bicycle (23 percent) or matatu/bus (33 percent; Table VII).

For pricing purposes, finfish were divided into five categories, namely small (< 20 cm), medium (20-50 cm) and large fish (> 50 cm) with the medium and large fish

Table IV.
Fish trading
specialization by male
(*n* = 93) and female
(*n* = 93) traders as a
percent of total in each
column

	Male	Female	Total
<i>Multiple response</i>			
No specialization	8.6	3.2	5.9
Small fish (< 20 cm)	43.0	67.7	55.4
Medium fish	64.5	68.8	66.7
Large fish	61.3	8.6	35.0
Crustaceans	17.2	6.5	11.8
Octopus	19.4	8.6	14.0
Other	1.1	0.0	0.5

Note: Traders could list more than one specialty (multiple responses) and therefore the total adds up to more than 100 percent

Table V.
The number of years
male (*n* = 93) and female
(*n* = 93) traders have
been at their trade, the
number of landing sites
that they frequent to
purchase their fish and
the number of fishers
from whom they
purchase the fish along
the Kenyan coast of
Malindi and Kilifi
districts

	Male	Female	Total
<i>Number of trading years</i>			
0-2	18.3	43.0	30.6
3-4	21.5	19.4	20.4
5-9	21.5	23.7	22.6
10+	38.7	14.0	26.3
<i>Number of landing sites</i>			
1	48.4	75.3	61.8
2	10.8	9.7	10.2
3	16.1	12.9	14.5
4	11.8	2.2	7.0
5-9	8.6	0.0	4.3
10+	4.3	0.0	2.2
<i>Number of fisher suppliers</i>			
1-2	11.1	28.7	19.8
3-5	21.1	37.9	29.4
6-10	31.1	26.4	28.8
11+	36.7	6.9	22.0

Fish traders on
the Kenyan coast

	Male	Female	Total
<i>Incentives</i>			
None	41.3	87.0	64.1
Financial assistance	28.3	2.2	15.2
Small gear	5.4	0.0	2.7
Foodstuffs	8.7	9.8	9.2
Combinations and other	16.3	1.1	8.7
<i>Fish cleaning</i>			
None	30.0	6.6	18.2
Gutting	68.9	91.2	80.1
Other	1.1	2.2	1.7
<i>Processing</i>			
None	35.5	14.0	24.7
Cooling	38.7	1.1	19.9
Drying	10.8	1.1	5.9
Frying	15.1	83.9	49.5

Note: Values are in percent

Table VI.
Incentives offered to
fishers by male ($n = 92$)
and female ($n = 92$) fish
traders and whether the
traders clean and process
their fish along the
Kenyan coast of Malindi
and Kilifi districts

	Male	Female	Total
<i>Location</i>			
On-site	2.2	1.1	1.6
NV	25.8	32.3	29.0
NV on tarmac road	15.1	39.8	27.4
Urban centres	48.4	11.9	30.1
Other	8.6	15.1	11.8
<i>Transportation</i>			
Foot	7.5	79.3	43.2
Bicycle	40.9	4.4	22.7
Matatu/bus	49.5	16.3	33.0
Private car	2.1	0.0	1.1

Note: Values are in percent of total within each column

Table VII.
Location where male
($n = 93$) and female
($n = 93$) sell their fish and
means of transporting the
fish

being further divided into grade A and grade B, depending on the degree of freshness. Fixed prices for the five categories were generally accepted with possible differences among landing sites (depending on accessibility and transport costs). Fishers and traders rarely argued about the price for a certain category of fish, but arguments did occur about the size classification of the catch and between grades A and B. The buying and selling prices reported by the traders for the five categories of fish were recorded. On average, there was a price difference of Ksh five per kg among the different categories of fish, which started at about Ksh 60 for the small fish to Ksh 80 for the large grade A fish (Table VIII).

Essentially, prices were decided by two factors, fish size and location of the landing site, but breakdown by site revealed some variation (Table IX). Notable was the higher prices that were paid for large fish in Ngomeni and Takaungu compared with the other sites. It is possible that this was caused by large fish such as tuna and sailfish at these

sites that were bought up directly for gourmet markets. Such fish were mostly absent in Mida and Kilifi and, if caught, not easily sold. In Malindi, the price of large fish was depressed but the market was distorted by the many sport vessels that landed their catch there. At the other end, the prices for small fish in Takaungu were considerably below the average of the other sites and this was related to the large seasonal catches of ribbon fish, a non-preferred fish.

Table X lists the average prices for catch categories and for landing sites together with the selling margins (the latter is expressed as a percentage of the buying price). The tracts that were difficult to access had selling margins of about 40 percent. The selling margins for the Kilifi coastal tract, near Kilifi town, where transport costs were less, were indeed lower (26 percent). Surprisingly, this was not the case at the sites near Malindi town and it appears that the market was functioning irregularly with producer prices being kept low and selling margins high. The selling margins for the different fish categories showed a rise from 32 to 40 percent.

Table VIII.
Buying and selling prices
(Ksh per kg) and margin
of the different fish
categories for male and
female traders along the
Kenyan coast of Malindi
and Kilifi districts

	Buy	Male Sell	Margin	Buy	Female Sell	Margin	Buy	Total Sell	Margin
Small fish (<20 cm)	68	87	19	58	79	21	62	82	20
Medium fish B (20-50)	66	93	27	66	88	22	66	91	25
Medium fish A (20-50)	71	97	26	69	92	23	70	94	24
Large fish B (50+)	76	108	32	69	90	21	75	105	30
Large fish A (50+)	81	112	31	69	90	21	80	110	30
Octopus	56	90	34	53	92	39	56	91	35
Crab	130	200	70	0	0	0	130	200	70
Prawns	263	333	70	0	0	0	234	309	75
Lobster	367	484	117	0	0	0	370	487	117

Table IX.
Average buying price
(Ksh per kg) of finfish
categories in the different
coastal tracts along the
Kenyan coast of Malindi
and Kilifi districts

	<i>n</i>	Ngomeni	Malindi	Mida	Kilifi	Takaungu
Small fish	102	66.0	69.2	62.5	58.9	44.4
Medium (B)	103	54.6	65.3	67.8	75.7	55.0
Medium (A)	120	57.3	71.3	67.7	77.4	66.6
Large (B)	62	90.5	68.3	72.3	80.0	65.0
Large (A)	69	94.6	70.0	72.3	85.0	93.3

Table X.
Average buying price
and selling margin (Ksh)
of finfish according to
coastal tract and fish
category along the
Kenyan coast of Malindi
and Kilifi districts

Coastal tract			Finfish category		
	Buying price	Selling margin		Buying price	Selling margin
Ngomeni	72.6	+40	Small fish	62.2	+32
Malindi	68.8	+41	Medium (B)	65.9	+37
Mida	68.5	+40	Medium (A)	70.0	+35
Kilifi	75.4	+26	Large (B)	75.2	+40
Takaungu	64.9	+42	Large (A)	80.0	+37

Gender differences

The trader population consists of half male and half female, with profound differences between genders (Table XI). Male traders were from mixed ethnic backgrounds. Many frequented more than one landing site and bought from an average of 11 fishers. They bought all types of fish, from small- to large-sized fish as well as specialties, and typically bought more than 20 kg, if not more than 50 kg. Occasionally they offered incentives to the fishers. Generally, they did not process the fish but some had access to a cooler or freezer. They sold in various places including nearby villages and urban centres and they transported the produce by mechanized vehicle. Half of the men were small-scale traders, more than a third operated as intermediaries and 10 percent were large-scale traders.

Women operated almost exclusively as small-scale traders or fishmongers who bought and processed mostly small fish and sold locally. They were typically of *Mijikenda* origin and had not been in the trade for very long. They frequented only one landing site and dealt with a few fishers that supplied them. They typically bought <20 kg per day, the majority even <10 kg per day. They did not offer additional incentives to fishers except occasional foodstuffs. Nearly all women traders fried the fish they purchased and did not use any other means of preservation or storage. The next selling location was either the NV or the NV on the tarmac road where the fish was taken by foot.

For male traders, the fish trade was their major activity and 50 percent reported that they had no additional economic activities. The fish trade was the only activity for 25 percent of the female traders while more than 50 percent reported they also did farming. In line with the description so far, the average trading income for female traders was about Ksh 800 per week, with 70 percent making less than Ksh 1,000 per week. For male traders, the average income was close to Ksh 1,700 per week; half the male traders earned more than Ksh 1,500 per week and 20 percent earned more than Ksh 2,500 per week (Table XII).

Discussion

All artisanal fishers in Malindi and Kilifi districts of the Kenya coast who went out to sea on boats were men. In addition, all fishers working from the shore by wading into the water and pulling nets back onto the land were men, although some women helped check the nets once they were pulled onto the land. However, about half the fish traders

	Male traders	Female traders
Number of landing sites frequented	Two or more	One only
Average number of fisher suppliers	11.1	5.1
Type of fish specialisation	All types	Small and medium fish
Daily fish purchases	50 kg or more	9 kg or less
Additional incentives	Yes	No
Fish processing	No	Frying
Fish storage	Yes	No
Nearest sales location	Rural and urban	Rural
Mode of transport	Mechanized	Foot
Average trade income (Ksh per week) (s.d.)	1,693 (1,001)	795 (674)

Table XI.
Typical characteristics of
male and female traders
in Malindi and
Kilifi districts of the
Kenya coast

were women. This pattern of men dominating the fish catch and women active in the processing and trading of the fish is common in most artisanal fisheries (Ochiewo, 2004). In fact, in many artisanal fisheries, the percentage of women traders are much higher than in this study and in some areas may even approach 100 percent (Bennett, 2005; Abbott *et al.*, 2007). Women could be active in this field for several reasons as pointed out by Béné (2006), namely, relatively low investments are required for small-scale processing and/or trading for local markets, operational costs are generally low, great physical strength is not required and it can be undertaken by relatively unskilled labour. These characteristics allow a large number of people from the lowest strata of the community who lack education, literacy and financial capital, especially women, to engage in this activity.

The proportion of women in the study was lower than that estimated by Medard *et al.* (2000) who reported that 70-80 percent of fish trade workers in the artisanal fish trade in Kenya, Tanzania and Uganda were women. The difference between the two reports could be due to ethnic considerations. Close to 97 percent of the women traders in our study area were *Mijikenda*, while only 2.2 percent were *Bajun* and none were *Swahili*/Arab. The latter two groups are Muslims and, traditionally, Muslim women cannot interact with men. Thus, these women could not be traders which, consequently, reduced the number of women traders.

An important constraint to fish traders is having a steady supply of fish all year (Abbott *et al.*, 2007). On the Kenyan coast, this is particularly true during the *kusi* season when less fish is available. In some areas in Africa, fishing households are also heavily involved in selling the fish, and the vendors start by selling the family's catch. This was reported to occur in the Upper Zambezi floodplains in Namibia (Abbott *et al.*, 2007). However, in our study, fish trading was not an important activity in the fisher households; among crewmembers, only 15 percent of the wives were engaged in fish trading or processing.

To ensure a fish supply, many traders enter into some agreement with the fishers. The traders offer incentives, mainly in the form of financial assistance, small gear and foodstuffs to the fishers. Such arrangements with fishers, mainly in the low season, were made by 35.9 percent of the traders, but there were large gender differences. About 58.7 percent of male traders offered incentives, mainly financial assistance (28.3 percent), but only 13 percent of females offered assistance. The most prominent

Table XII.
Incomes (Ksh per week)
among male (*n* = 91) and
female (*n* = 82) fish
traders from fish trading
only along the Kenya
coast of Malindi and Kilifi
districts

	Male	Female	Total
To 399	0.0	23.2	11.0
400-599	9.9	25.6	17.3
600-699	5.5	9.8	7.5
700-999	6.6	11.0	8.7
1,000-1,299	17.6	14.6	16.2
1,300-1,999	23.1	3.7	13.9
2,000-2,499	16.5	8.5	12.7
2,500+	20.9	3.7	12.7
Means	1,693	795	1,268
(s.d.)	(1,001)	(674)	(970)

Note: Values represent the percentage in each income group

actors offering assistance were the *tajiris*, all men. They often owned the fishing boats and gear and provided soft loans to the fishers in times of financial hardships, which made the fishers very dependent on them. In return, the *tajiris* received a portion of the catch and had first refusal of the catch of certain fishers.

A disturbing, growing new phenomenon in the artisanal fish trade is fish-for-sex transactions. Most cases, 97 percent, have been reported in Africa, with 49 percent of the cases in East Africa. Here, women engage in sexual relationships with male fishers to secure their fish supply and, also, at times to buy fish for a cheaper price. In fact, some fishers would not sell to women unless they have sexual relationships with them. Most vulnerable to these pressures are the single and widowed women fish traders (Béné and Merten, 2008).

Earnings for fish traders

We questioned whether fish traders in the artisanal fisheries along the Kenyan coast earned enough money from only fish trading to support a household. Average income for the fish traders from only fish trading was Ksh 1,268 per week. With an average household size of 7.7 people (Hoorweg *et al.*, 2009a), monthly earnings from fish trading amounted to Ksh 714 per person, which was substantially lower than the national poverty line set at Ksh 1,239[3]. In order for a household of 7.7 people to reach the poverty line, a weekly income of Ksh 2,201 was needed. This was achieved by about 20.3 percent[4] of the traders, that is, 79.7 percent of the household were below the poverty line.

However, there was a large difference between male and female traders. Men earned Ksh 1,693 per week, with half the traders earning more than Ksh 1,500 per week and 20 percent earning more than Ksh 2,500 per week. In contrast, the average earning of women, about Ksh 795 per week, was about half that of men, with 70 percent making less than Ksh 1,000 per week. With the same assumptions as above, monthly earnings from only fish trading amounted to Ksh 953 per person for male traders and Ksh 447 per person for female traders. About 30.8 percent of the male traders' households reached the poverty line but only 8.8 percent of the female traders did so.

Livelihood diversification, a widespread survival strategy among rural households in Africa (Ellis, 2000), was a common occurrence for both the traders and fishers. Diversification is generally expected to improve household resources, if not resulting in increases in income then at least resulting in a wider income spread, although other researches consider specialization as the more effective way of improving income. Two types of diversification were identified: earner diversification when the fish trader (in this study) had another income and activity diversification when someone else in the household had an income (Hoorweg *et al.*, 2009b). For all traders, about 50 percent of the men and 75 percent of the women had an income other than from fish trading. Farming income was the most common, but self-employment and wage labour were also mentioned. In comparison, 51 percent of fishers had earnings other than from fishing (activity diversification). Here too, it was mainly from farming and to a lesser extent from self-employment and wage labour. In 39 percent of the households, someone else besides the fisher earned money (earner diversification) and in 19 percent of the households, both the fisher and someone else in the household earned money other than from fishing.

The income for fishers from fishing was Ksh 1,439 (Hoorweg *et al.*, 2009a) and from all earnings was Ksh 1,696 per week. Earnings for the fisher household were Ksh 1,952 per week. Therefore, the percentage of earnings from fishing was approximately 85 percent of the fishers' earnings and 74 percent of the household income. If we assume the same proportion of earnings from fish trading as the fishers earned from fishing, then the earnings of the traders would be Ksh 1,492 per week from only fish trading and Ksh 1,714 per week per household. This would amount to Ksh 840 per person per month when using only the fisher earnings and Ksh 965 per person per month when using the household earnings. About 72.6 percent of the traders' households would be considered at or below the poverty line.

However, the scene changes dramatically when calculating male and female fish traders separately. For men, about 51 percent reported that they had additional economic activities, a similar percentage to that of the fishers. If we assume the same proportion of earnings from fish trading as the fishers earned from fishing, then the earnings would be Ksh 1,992 per week from fish trading and Ksh 2,288 per week per household. This would amount to Ksh 1,121 per person per month when using only the fish trading earnings and Ksh 1,288 per person per month when using the household earnings. It was calculated that only 46 percent of households from male traders fell below the poverty line. This is far better than the 59 percent for fisher households and 64 percent for the rural population in Coast Province overall (Hoorweg *et al.*, 2009a).

For 25 percent of the female traders, fish trade was their only activity, which is considerably less than the male fish traders. Again, if we assume that female earnings from fish trading were in the same proportion of the total earnings as male traders and taking into account that nearly two times as many females showed earner diversification than did the males, then earnings for the females would be Ksh 1,049 per week per trader and Ksh 1,351 per week per household. This would amount to Ksh 590 per person per month in the household when using only the trader earnings and Ksh 760 per person per month when using the household earnings. About 85 percent of the households would be at or below the poverty line. Although women easily enter fish trading, their activities are usually small scale and their incomes less than their male counterparts. They suffer constraints such as lack of credit and training, inadequate markets and storage and transport problems that deny them access to distant markets.

Notes

1. Recent estimates show that Coast Province remains with a high poverty rate – about 58 percent (Kenya, 2005).
2. A *tajiri* is a person who leases vessels and gear to fishers for a payment of between 20 and 50 percent of the daily catch; the remainder is shared among the fishers operating the vessel (Glaesel, 1997). The *tajiri* also sustains the fisheries by providing soft loans to fishers in times of financial hardships, making them highly dependent on him. In Malindi and Mida, the role of the *tajiri* is small. In Malindi the diminished role of the *tajiri* has to do with the presence of a co-operative, which provides financial support to the fishers through a loan system. For Mida, the diminished role of the *tajiri* is probably due to the small scale of the fishing activities there. In Ngomeni, Kilifi and Takaungu the *tajiri* plays a more important role, particularly in Ngomeni where in 20 percent of the cases the *tajiri* supplied gear. This probably is because of the larger vessels and heavier gear used in fishing at this part of the coast, which are more costly.

3. At the time of the study; the poverty line was set at Ksh 1,239 per person per month (~US\$19.8 per month or ~US\$0.66 per day) in rural areas and Ksh 2,648 per person per month in urban areas (Kenya, 2001).
4. Calculation was as follows (using Table XII for earnings and 2201 as poverty line) 12.7 percent of male traders earned Ksh 2500 or more – therefore, all above poverty line. 12.7 percent earned Ksh 2,000-2,499. Of these, 7.6 percent were above the poverty line as follows: $(2,499 - 2,201) / (2,499 - 2,000) \times 12.7 \text{ percent} = 7.6 \text{ percent}$; $12.7 \text{ percent} + 7.6 \text{ percent} = 20.3 \text{ percent}$.

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